Design and Analysis of Algorithms a.k.a. Analysis of Algorithms
Special Section 2024 (Summer 24)
Quiz / Assignment 01 - 08-07-2024 Due on: 11-07-24

Answer, to the point, for the following:

[1. 5 marks] Define algorithm (as taught in class)

[2. 5 marks] Name five Properties of Algorithms.

[3. 4 marks] What are the four ways to represent algorithms.

[4. 5 marks] Write the first 10 figures of the ternary counting with zero as 0, one as 1 and three as 3.

[5. 5 marks] Write the first 10 figures of the ternary counting with zero as 8, one as 9 and three as 6.

[6. 5 marks] What is the price of eBook of the textbook used in the course?

[7. 2 marks] Is eTextbook available on rental basis?

[8. 2 marks] How many mistakes (Errata) is officially reported?

[9. 3 marks] In the preface of the book, what guidance is given to the students?

[10. 4 marks] Which two sorting techniques are referred to in the topic Algorithms as technology (Efficiency)?

[11. 10 marks] If, in discrete (0/1) knapsack problem, the capacity of knapsack is 26, what is the optimal set from the following 10 items descriptions.

```
weight[10]   = { 15,  12,  10,   9,   8,   7,  5,  4,  3,  1};
value[10]    = {210, 220, 180, 120, 160, 170, 90, 40, 60, 10};
```

[12. 20 marks] Consider for some instances of a 0/1 knapsack problem having completely distinct items, a greedy algorithm can solve the problem. Write the algorithm with greed being ratio value/weight for each item.

[13. 10 marks] Understand the following recursive algorithm and make corrections (if any) in it. Italic lines are comments.

```
def SelectionSort(ar, n):
# ar is array/list of size n
    if n == 1:
        Return
    mxloc = Position(ar, n)
    # position return the index of maximum value in array of size n
    Swap(ar, n-1, mxloc)
    # swap the values in array ar at indices n-1 and mxloc
    SelectionSort(ar, n-1)
```

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[14. 10 marks] Write the algorithms (iterative or recursive) for position and swap in above example.

[15. 20 marks] Write a recursive algorithm to return the minimum of all the elements of an array or list using two recursive calls for left and right halves of the array or lists.

[16. 20 marks] Like sets have subsets, lists can have sublists. A sublist contains zero (for empty sublist) or more elements from the list from random places but maintaining order of the original position of elements (even repeating). Design and write an algorithm to return the longest common sublist of the two lists. In case multiple sublists of same maximum size are there in an instance (as in figure below) return any of them.


```
def LongestSubList(ar1, n1, ar2, n2):
# ar1 and ar2 is array/list of sizes n1 and n2 respectively
```

[17. 10 marks] Consider integers have too many digits and they are stored as lists/arrays of digits. Write algorithm to return a list/array containing the sum of two long integers.

```
def Sum(ar1, n1, ar2, n2):
# ar1 and ar2 is array/list of sizes n1 and n2 respectively
```

[18. 20 marks] Write loop invariant for the following algorithm. Also, using it, prove that the algorithms is correct.

[19. 10 marks] Prove that algorithm given in Q-13 is correct.

```
def LinearSearch(x, ar, n):
# ar is array/list of size n, x is value to search in array ar
# return true if x in array somewhere (maybe multiple times) else return false
    ans = false
    j = 1
    while j < n:
        if ar[j] == x:
            ans   true
    return ans
```

[20. 10 marks] Write iterative algorithm to return sum of its elements. Prove it is correct.